

A Short Guide to Proposition C.6

The hybrid Gaussian flat-prior upper bound

Abstract

Proposition C.6 is the converse for the hybrid Gaussian experiment containing both the classical spectral variable and the quantum orbital modes. It shows that even a channel coupling these two components cannot beat the product value $B_{\text{univ}}(\gamma, p) = B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p)$. The proof uses a classical Gaussian witness and a compact quantum witness, extracts a covariant quantum limit and a translation-covariant scalar limit, and bounds them separately. This note organizes the argument into its two parallel parts and explains how they recombine into the sharp fidelity bound.

1 The hybrid Gaussian experiment

The classical space is

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad \dim \mathbf{H} = d - 1,$$

with positive covariance Σ_p . Write

$$f_1 = \varphi_{0, \Sigma_p}, \quad f_\gamma = \varphi_{0, \gamma \Sigma_p}$$

for the centered Gaussian densities. The orbital component lives on the multimode Fock space $\mathcal{F}_{\text{orb}}$ and has centered thermal state τ .

At parameter $\xi = (h, z) \in \mathbf{H} \times \mathbb{C}^{r_d}$, the input and ideal target are

$$\begin{aligned} \Theta_{h,z}^{\text{in}} &= \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z, \\ \Theta_{h,z}^{\text{tar}} &= \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z. \end{aligned}$$

The target amplifies both means by $\sqrt{\gamma}$ without increasing either the classical covariance or the centered thermal noise. A physical procedure cannot do this perfectly.

The optimal product procedure dilates the classical observation by $\sqrt{\gamma}$ and applies the product quantum-limited amplifier. Its two losses are

$$B_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \tag{1}$$

$$B_{\text{orb}}(\gamma, p) = B(\tau^{(\gamma)}, \tau), \tag{2}$$

and root fidelity multiplies across the classical and quantum factors. Proposition C.6 proves that a joint hybrid channel cannot improve on their product.

2 Statement of Proposition C.6

Let Ω_L be the centered cube in $\mathbb{H} \times \mathbb{C}^{r_d}$ used in the manuscript.

Proposition C.6 (rephrased). *For the hybrid Gaussian input and target families,*

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} B\left(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}\right) dh dz \leq B_{\text{univ}}(\gamma, p),$$

where $B_{\text{univ}}(\gamma, p) = B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p)$ and the supremum is over all hybrid channels, including channels that couple the classical and quantum components.

3 The two witnesses

3.1 Classical witness

Define the square-root likelihood ratio

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}} = \gamma^{(d-1)/4} \exp\left[-\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle\right]. \quad (3)$$

It is bounded, strictly positive, and belongs to $C_0(\mathbb{H})$. Moreover,

$$\int_{\mathbb{H}} \frac{f_1(y)}{\chi(y)} dy = \int_{\mathbb{H}} \sqrt{f_1(y)f_\gamma(y)} dy = B_{\text{cl}}(\gamma, d). \quad (4)$$

The fact that χ vanishes at infinity is as important as the Gaussian identity: it makes integration against χ continuous for weak-* limits of finite measures.

3.2 Quantum witness

In the joint number basis of τ and the quantum-limited output $\tau^{(\gamma)}$, define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (5)$$

It is positive, injective, bounded, compact, and a product of decreasing photon-number weights. Its matching identities are

$$\text{Tr}(\tau W_\gamma^{-1}) = \text{Tr}(\tau^{(\gamma)} W_\gamma) = B_{\text{orb}}(\gamma, p). \quad (6)$$

Lemma C.1 combines these witnesses. If $R(y)$ is a positive classical–quantum density, then

$$B(R, f_1 \tau)^2 \leq \left(\int_{\mathbb{H}} \chi(y) \text{Tr}[R(y) W_\gamma] dy \right) B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (7)$$

It remains to show that the first, channel-dependent factor is at most the same product $B_{\text{cl}} B_{\text{orb}}$.

4 Proof roadmap

The proof performs the following reductions:

$$\begin{array}{ccc}
& \text{average fidelity over a large parameter cube} & \\
& \Downarrow & \\
& \text{fidelity at the origin for a Følner-averaged channel} & \\
& \Downarrow & \\
& \text{one weighted moment } M_j & \\
& \Downarrow & \\
\text{quantum compact limit } \Gamma^\chi & & \text{classical scalar limit } \mathcal{R} \\
\text{Lemma C.2} & & \text{Lemma C.5} \\
& \Downarrow & \\
M_j \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p) & & \\
& \Downarrow & \\
B \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). & &
\end{array}$$

The two limits come from the same averaged channels but retain different information. The compact quantum limit controls the W_γ moment; the scalar limit controls how much classically weighted trace that quantum map can have.

5 Proof walkthrough

Step 1: Følner averaging

For $\xi = (h, z)$, let

$$\begin{aligned}
(\mathcal{U}_\xi^{\text{in}} R)(x) &= D(z)R(x-h)D(z)^*, \\
(\mathcal{U}_\xi^{\text{out}} R)(y) &= D(\sqrt{\gamma}z)R(y-\sqrt{\gamma}h)D(\sqrt{\gamma}z)^*.
\end{aligned}$$

Then

$$\Theta_\xi^{\text{in}} = \mathcal{U}_\xi^{\text{in}}(f_1\tau), \quad \Theta_\xi^{\text{tar}} = \mathcal{U}_\xi^{\text{out}}(f_1\tau).$$

For a hybrid channel Λ , define

$$\bar{\Lambda}_L = \frac{1}{|\Omega_L|} \int_{\Omega_L} \mathcal{U}_{-\xi}^{\text{out}} \circ \Lambda \circ \mathcal{U}_\xi^{\text{in}} d\xi. \quad (8)$$

Joint concavity and invariance of root fidelity give

$$\frac{1}{|\Omega_L|} \int_{\Omega_L} B(\Lambda(\Theta_\xi^{\text{in}}), \Theta_\xi^{\text{tar}}) d\xi \leq B(\bar{\Lambda}_L(f_1\tau), f_1\tau). \quad (9)$$

Choose $L_j \rightarrow \infty$ and channels approaching the limsup, and denote their averages by Λ_j . Translating the averaging cube shows that for each fixed ξ ,

$$\|\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) - \mathcal{U}_\xi^{\text{out}} \Lambda_j(R)\|_1 \leq \frac{|(\Omega_{L_j} + \xi) \triangle \Omega_{L_j}|}{|\Omega_{L_j}|} \|R\|_1 \rightarrow 0. \quad (10)$$

Step 2: reduce fidelity to one moment

Put

$$R_j = \Lambda_j(f_1 \tau), \quad M_j = \int_{\mathbf{H}} \chi(y) \operatorname{Tr}[R_j(y) W_\gamma] dy.$$

The weighted hybrid inequality (7) gives

$$B(R_j, f_1 \tau)^2 \leq M_j B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (11)$$

The rest of the proof is devoted to the upper bound

$$\limsup_j M_j \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (12)$$

Step 3: extract the weighted quantum map

For a trace-class operator X on $\mathcal{F}_{\text{orb}}$, define

$$\Gamma_j^\chi(X) = \int_{\mathbf{H}} \chi(y) [\Lambda_j(f_1 X)](y) dy. \quad (13)$$

This map feeds Λ_j a fixed classical Gaussian with quantum input X , then averages the quantum output with the classical weight χ . It is completely positive and satisfies

$$\operatorname{Tr} \Gamma_j^\chi(X) \leq \|\chi\|_\infty \operatorname{Tr} X \quad (X \geq 0).$$

Notice that $\|\chi\|_\infty = \gamma^{(d-1)/4}$ may exceed one. Hence this initial estimate does *not* make Γ_j^χ trace nonincreasing.

Taking $h = 0$ in (10) shows that these maps become displacement covariant in the quantum variable. Lemma C.2 therefore gives, after taking a subsequence, a completely positive exactly covariant map Γ^χ with

$$\operatorname{Tr}[\Gamma_j^\chi(X) K] \longrightarrow \operatorname{Tr}[\Gamma^\chi(X) K] \quad (K \text{ compact}). \quad (14)$$

Because W_γ is compact,

$$M_j = \operatorname{Tr}[\Gamma_j^\chi(\tau) W_\gamma] \longrightarrow \operatorname{Tr}[\Gamma^\chi(\tau) W_\gamma]. \quad (15)$$

To apply the least-noise moment to Γ^χ , the proof still has to improve its trace bound from $\|\chi\|_\infty$ to one. This is the purpose of the classical limit.

Step 4: extract the scalar trace kernel

For $u \in L^1(\mathbf{H})$, let $\mathcal{R}_j(u)$ be the classical trace measure of $\Lambda_j(u\tau)$:

$$\mathcal{R}_j(u)(E) = \int_E \operatorname{Tr}[\Lambda_j(u\tau)(y)] dy. \quad (16)$$

Since Λ_j is a channel, $\mathcal{R}_j$ is a positive contraction from $L^1(\mathbf{H})$ to the finite Radon measures.

A diagonal extraction on countable dense subsets of $L^1(\mathbf{H})$ and $C_0(\mathbf{H})$ gives a positive contraction $\mathcal{R}$ such that

$$\mathcal{R}_j(u) \xrightarrow{w^*} \mathcal{R}(u) \quad (u \in L^1(\mathbf{H})).$$

Taking $z = 0$ in (10) and passing to this limit yields

$$\mathcal{R}(u(\cdot - h))(E) = \mathcal{R}(u)(E - \sqrt{\gamma}h). \quad (17)$$

Lemma C.5 identifies this covariant scalar kernel: there is a finite positive measure μ , with $\mu(\mathbf{H}) \leq 1$, such that

$$\mathcal{R}(u)(E) = \int_{\mathbf{H}} u(x) \mu(E - \sqrt{\gamma}x) dx. \quad (18)$$

Step 5: bound the trace of the weighted quantum limit

Let $P_N \uparrow I$ be finite-rank projections on $\mathcal{F}_{\text{orb}}$. Positivity and (14) give

$$\begin{aligned} \text{Tr } \Gamma^\chi(\tau) &= \sup_N \text{Tr}[\Gamma^\chi(\tau)P_N] \\ &\leq \liminf_j \text{Tr } \Gamma_j^\chi(\tau). \end{aligned} \quad (19)$$

By (13) and (16),

$$\text{Tr } \Gamma_j^\chi(\tau) = \int_{\mathbf{H}} \chi(y) \mathcal{R}_j(f_1)(dy).$$

Because $\chi \in C_0(\mathbf{H})$, weak-* convergence of the scalar measures passes this integral to the limit. Hence

$$\text{Tr } \Gamma^\chi(\tau) \leq \int_{\mathbf{H}} \chi(y) \mathcal{R}(f_1)(dy). \quad (20)$$

Use the convolution representation (18). If $X \sim \mathbf{N}_{0, \Sigma_p}$, then $\sqrt{\gamma}X$ has density f_γ , so

$$\int_{\mathbf{H}} \chi(y) \mathcal{R}(f_1)(dy) = \int_{\mathbf{H}} c_\gamma(r) \mu(dr), \quad (21)$$

$$c_\gamma(r) = \int_{\mathbf{H}} \chi(y) f_\gamma(y-r) dy. \quad (22)$$

Completing the Gaussian square gives

$$c_\gamma(r) = B_{\text{cl}}(\gamma, d) \exp\left[-\frac{\gamma-1}{2\gamma(\gamma+1)} \langle r, \Sigma_p^{-1}r \rangle\right] \leq B_{\text{cl}}(\gamma, d). \quad (23)$$

The maximum occurs at zero added classical noise. Since $\mu(\mathbf{H}) \leq 1$, equations (20)–(23) yield

$$\text{Tr } \Gamma^\chi(\tau) \leq B_{\text{cl}}(\gamma, d). \quad (24)$$

This estimate is first proved only at τ . Exact displacement covariance makes it global. Indeed, the positive functional $X \mapsto \text{Tr } \Gamma^\chi(X)$ is represented by a bounded positive operator A . Covariance implies that A commutes with every multimode Weyl displacement, so irreducibility gives $A = cI$. Since $\text{Tr } \tau = 1$,

$$c = \text{Tr } \Gamma^\chi(\tau) \leq B_{\text{cl}}(\gamma, d) \leq 1.$$

Therefore Γ^χ is trace nonincreasing and Proposition C.3 applies.

Step 6: apply the quantum least-noise bound

Using (15), Proposition C.3, and (6),

$$\begin{aligned} \lim_j M_j &= \text{Tr}[\Gamma^\chi(\tau)W_\gamma] \\ &\leq \text{Tr } \Gamma^\chi(\tau) \text{Tr}(\tau^{(\gamma)}W_\gamma) \\ &\leq B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p). \end{aligned}$$

This proves (12). Substitution into (11) gives

$$\limsup_j B(R_j, f_1\tau)^2 \leq (B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p))^2.$$

Taking square roots and using the Følner concavity bound (9) proves Proposition C.6.

6 Why the classical and quantum factors multiply

The proof does not assume that the competing channel factors. Instead, factorization emerges at the level of witnesses:

- χ is the square-root likelihood ratio for the two classical Gaussian covariances;
- W_γ is the square-root likelihood-ratio observable for the two thermal laws;
- Lemma C.1 separates the channel-dependent moment from the two fixed inverse-witness integrals;
- Lemma C.5 bounds the trace available to the quantum moment by the classical affinity B_{cl} ;
- Proposition C.3 bounds the normalized quantum moment by the orbital affinity B_{orb} .

Thus any classical–quantum coupling is already included in the single moment M_j , yet the covariance constraints force that moment to pay both losses.

7 Common points of confusion

1. The compact-limit lemma alone gives $\text{Tr } \Gamma^\chi(X) \leq \|\chi\|_\infty \text{Tr } X$, which is not enough for Proposition C.3. The scalar convolution calculation is what improves the coefficient to $B_{\text{cl}} \leq 1$.
2. Two different compactness arguments are used. Lemma C.2 treats quantum maps through compact operators, while the diagonal extraction of $\mathcal{R}_j$ treats classical finite measures through $C_0(\mathbf{H})$.
3. Compactness of W_γ passes the quantum moment to the limit; membership $\chi \in C_0(\mathbf{H})$ passes the classical weighted trace to the limit. These are parallel requirements.
4. The noise measure μ need not be normalized or absolutely continuous. The estimate uses only positivity and $\mu(\mathbf{H}) \leq 1$.

8 Role in the paper

Proposition C.6 is the upper-bound half of the hybrid Gaussian flat-prior theorem. The product of classical dilation and quantum-limited amplification attains the same value, so

$$B_{\text{univ}}(\gamma, p) = B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p)$$

is sharp. LAN fidelity transfer then carries this Gaussian converse back to the unknown-spectrum cloning problem and produces the classical penalty absent from known-spectrum cloning.